

Navya Jain

Boston, MA | iamjainnavya@gmail.com | +18573899933 | [Website](#) | [LinkedIn](#)

WORK EXPERIENCE

Ritchie Bros (acquired **Veritread** in March 2026), **Boston (Remote)**

July 2024 - Present

AI/ML Engineer (Oct 2025-Present)

- Led development of an **AI bidding agent** supporting concurrent users and session memory with **AWS Bedrock AgentCore**, **Strands SDK**, **Claude Sonnet**, and **MCP** tools used by admins, reducing manual bid processing time by 85%
- Owned **end-to-end machine learning pipelines** with **PyTorch** and **scikit-learn** (regressors) to generate price estimations for transporting heavy hauls, increasing quarterly revenue by 20%
- Engineered **100+ time-series features** and **cached** recent quotes for a **LightGBM confidence model** to **score price prediction accuracy**, enabling stakeholders to set risk thresholds and guardrails on automated quotes
- Added an **LLM-based plausibility** guardrail (Claude) to route flagged prices with the reasoning so that humans can accept or override before a price ships
- Developed a **Thompson sampling contextual bandit** for the pricing model from an 8-week, 5-arm randomized markup experiment, lifting margin per quote 12% vs. a permanent fixed-markup holdout
- Ran an online **A/B test** to measure the incremental impact of a new pricing model, establishing a statistically significant 37% lift in acceptance with no revenue degradation, where the incrementality read-out drove the go/no-go launch decision
- Built an **evaluation harness** for the agent using 50 time-split bidding scenarios to regression-test 15+ prompt versions for tool selection, failure handling, and multi-turn behavior
- Built a **recommendation engine** that suggested top 10 carriers for a load based on prior activity, achieving 0.87 Recall@10 and a 64% lift over a popularity baseline using **matrix-factorization embeddings** and **weighted ranking**

Software Engineer (Machine Learning) (July 2024 - Sept 2025)

- Designed tools for price forecasting using **gradient boosting** methods in **scikit-learn** for client-specific freight movement
- Deployed ML pricing models on **Amazon SageMaker AI** and made a REST API for generating predictions, pointing to an existing **EKS** (Kubernetes) service to handle 1M monthly requests
- Wrote **.NET 8** and **C# REST APIs** for internal services enabling reliable **SQL**-based data access on the backend with **JWTAuth**
- Conducted MLOps by automating model retraining with **EventBridge** and **Lambda**, maintaining observability across **CloudWatch** and **Datadog**, and shipping to dev/QA/prod via **Docker** and **CircleCI**

Realix.ai, Boston (Remote) (Part-time contract)

November 2024 - January 2025

Data Scientist

- Shipped a GPT-4 Turbo **RAG pipeline** (**LlamaIndex**, **MySQL**) that converts natural language into executable SQL and validated SQL accuracy across consistent SQL output across 15+ phrasings of the same queries
- Maintained **ETL pipelines** using Python and **PostgreSQL**, and made **Tableau** dashboards that improved planning efficiency

Boston University, Boston

January 2023 - January 2024

Head Teaching Assistant for Foundations of Data Science III

- Led discussion sessions and assembled assignments on probability, statistics, and data science, covering Bayesian statistics, optimization, and sampling with Python (pandas, numpy, scipy, scikit-learn) for 120 students

EDUCATION

M.S. Artificial Intelligence (AI) | Boston University, Boston, MA

September 2022 - January 2024

B.Tech Elec & Electronics Engg | Guru Gobind Singh Indraprastha University, Delhi

August 2018 - June 2022

PROJECTS

Fine-tuning VLM to Read Aviation Sectional Charts ([GitHub](#))

- Fine-tuned **Qwen3-VL-4B** with **QLoRA** on 50,000 auto-labeled FAA chart crops to extract airspace, terrain, and frequencies
- Built a data pipeline linking chart images with official FAA records, achieving 89% field-level accuracy
- Evaluated **visual grounding**, **symbol recognition**, **OCR**, and structured outputs across 5,000 held-out chart crops

Semantic Book Recommendation with Two-Tower CF Architecture ([GitHub](#))

- Created an **agentic** book recommender in **PyTorch** with over 2M books and 250,000 user ratings to translate natural language preferences into personalized results using **two-tower retrieval** and **LLM re-ranking**
- Combined **neural collaborative filtering** with BM25 and dense semantic search (**sentence-transformers**) over review passages to surface books matching user taste and emotional intent beyond keyword match
- Built reproducible PySpark and Airflow pipelines and deployed the system on **GCP**

Multimodal Vision-Language Model (PaliGemma) from Scratch

- Implemented a **PyTorch multimodal architecture** by implementing a **SigLIP Vision Transformer**, Gemma decoder, and a projection bridge, with custom RoPE, RMSNorm, attention, and KV caching

SKILLS

- **Programming Languages:** Python, SQL, TypeScript, C++
- **Machine Learning:** PyTorch, scikit-learn, LightGBM, MLflow, pandas, HuggingFace, TensorFlow, Sentence-Transformers
- **Agentic/gen AI:** Model Context Protocol (MCP), AWS Bedrock, LangChain, LlamaIndex, QLoRA, PEFT
- **Cloud, Data and MLOps:** AWS, Docker, CircleCI, Datadog, Git, SSMS, Postgres, Tableau, PySpark, Apache Airflow, Redis, Feast
- **Methods:** A/B testing, contextual bandits, two-tower models, time-series forecasting, agentic workflows, RAG, fine-tuning

OTHER EXPERIENCE

Nawaazish Crew(Charity), Delhi

February 2020 - January 2021

Co-founder and President

- Co-founded a nonprofit in India; led 45 volunteers and led COVID relief for 30,000 families, and partnered with organizations like Action for Autism and Robinhood Army to conduct data-driven drives